

1 spxtacular: A Chainable Python Library for Mass 2 Spectrum Processing

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Software

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5 Summary

6 Mass spectra record detected ions as peaks, each defined by a mass-to-charge ratio (m/z) and
7 an intensity. Extracting chemical evidence from these peaks may require denoising, isotope
8 deconvolution, charge assignment, neutral-mass calculation, fragment matching, and scoring.
9 Reimplementing these steps in individual analysis scripts can make workflows difficult to
10 reproduce and intermediate results difficult to inspect.

11 **spxtacular** organizes these operations as methods on a chainable Spectrum object. The object
12 keeps m/z and intensity values aligned with optional per-peak charge, ion mobility, and isotopic
13 profile scores. Metadata produced by each operation remains attached to the spectrum during
14 subsequent filtering, plotting, matching, and scoring.

```
neutral = (spec.denoise(method="mad")
            .deconvolute(charge_range=(1, 4), tolerance=15,
                        tolerance_type="ppm", min_score=0.5)
            .decharge())
```

15 **Figure 1** summarizes how readers, processing methods, interpretation, and outputs operate
16 through the same metadata-bearing spectrum object.

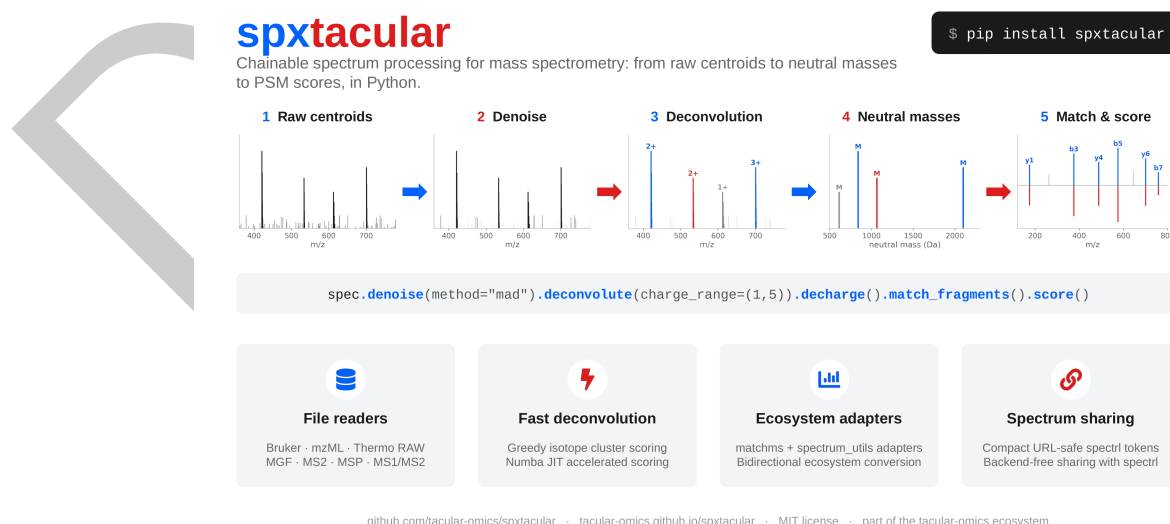


Figure 1: Overview of the spxtacular workflow. A chainable Spectrum pipeline transforms raw centroids through denoising, deconvolution, neutral-mass conversion, fragment matching, and scoring. File readers, accelerated deconvolution, interactive plots, and compact spectrum sharing support the workflow.

The core package depends on NumPy (Harris et al., 2020), pandas (McKinney, 2010), Plotly (Plotly Technologies Inc., 2015), peptacular (Garrett & Yates, 2026c), and paftacular (Garrett & Yates, 2026b). An optional spectrl integration (Garrett & Yates, 2026d) provides URL-safe spectrum encoding.

Statement of need

Existing Python tools cover important parts of this workflow. Pyteomics (Levitsky et al., 2019) emphasizes file access and proteomics utilities, whereas pyOpenMS (Röst et al., 2014) exposes the broader OpenMS data model and algorithms. During interactive method development, moving among tools can still require repeated data-model conversions and manual preservation of peak-level evidence.

spxtacular addresses this coordination problem by returning the result of deconvolution as another spectrum rather than a file or a set of detached arrays. The result can then be filtered, plotted, matched against fragments, or processed again in the same session. Charge and isotope scores remain attached to each peak as evidence for its assignment.

The intended users are mass spectrometry researchers, method developers, and software authors who need to inspect or transform spectra without committing to a complete analysis pipeline. Built-in isotope models support peptide, glycan, lipid, DNA, and RNA analytes, while custom average-composition models extend the same workflow to other chemical classes. A common interface supports exploratory analysis, reusable processing code, and integration into larger workflows.

State of the field

Several Python packages offer complementary capabilities. ms_deisotope (Klein et al., 2024) supports averagine-based deisotoping and charge-state deconvolution, while matchms (Huber et al., 2020) supports spectrum similarity and composable processing. spectrum_utils (Bittremieux et al., 2023) focuses on annotation and plotting, and pyOpenMS offers a broader collection of mass spectrometry algorithms.

spxtacular focuses on coordinating these tasks through a single spectrum model that retains reader metadata, deconvolution evidence, fragment assignments, scores, and visualization inputs. Adding this workflow to any one of the existing packages would have tied it to that package's data model and scope. Instead, adapters connect spxtacular to matchms and spectrum_utils while preserving its peak-level metadata and processing state.

Software design

The spectrum-processing code originated in peptacular. It was separated into spxtacular so that spectrum input and output, processing, visualization, and interoperability could evolve independently while peptacular remained focused on peptide chemistry and fragment generation.

The central design choice is a Spectrum object that keeps m/z and intensity arrays aligned with optional per-peak metadata. Its methods preserve compatible metadata and return a distinct spectrum by default. This behavior improves predictability and reproducibility at the cost of additional memory. Users can select an explicit in-place option when memory use is a concern. A SpectrumType value records whether the spectrum contains profile, centroid, or deconvoluted data and prevents invalid operation sequences such as neutral-mass conversion before deconvolution.

Visualizations are generated directly from the spectrum and its per-peak metadata. A Spectrum can therefore be displayed as a stick, mirror, or fragment-annotated plot and inspected as an

61 HTML peak table without an intermediate conversion.

62 Deconvolution and scoring

63 Isotope cluster detection follows a greedy strategy related to MS-Deconv (Liu et al., 2010)
 64 and THRASH (Horn et al., 2000). Aggregated nominal isotope distributions are calculated
 65 with a BRAIN-style Newton–Girard recurrence (Dittwald & Valkenburg, 2014) from an average
 66 elemental composition. Built-in models cover peptides, glycans, lipids, DNA, and RNA, and
 67 users can define custom compositions and isotope abundances. Envelope length expands
 68 adaptively until the predicted tail falls below the selected abundance threshold.

69 Beginning with the most intense unused peak, the algorithm evaluates every candidate charge
 70 state and aligns the seed with the predicted envelope apex or a contiguous near-apex isotope.
 71 It then matches outward in both directions using mass-error, abundance-fold, and optional
 72 ion-mobility gates. Each candidate is compared with the complete predicted envelope using
 73 the Bhattacharyya coefficient, so both missing predicted intensity and unexpected observed
 74 intensity reduce the score; missing peaks that should have been detectable receive an additional
 75 penalty. The best charge and alignment consume their matched peaks only after all candidates
 76 have been evaluated. This apex-first procedure can infer the monoisotopic m/z when the
 77 $A+0$ peak is not observed. If the best score falls below `min_score`, the seed is retained as a
 78 singleton while the other candidate peaks remain available for subsequent searches.

79 Assigned peaks retain their charge and `iso_score`, allowing users to inspect and filter
 80 deconvolution results without discarding the evidence behind individual assignments.

81 Readers and interoperability

82 Reader dispatches by path suffix to backends for Bruker timsTOF .d directories, mzML files
 83 (Martens et al., 2011), Thermo .raw files, and MGF, MS2, or MSP peak lists. The mzML
 84 and Bruker backends depend on `mzmlpy` (Garrett & Yates, 2026a) and `tdfpy` (Garrett &
 85 Yates, 2026e), respectively. Every backend yields the same `MsnSpectrum` type with available
 86 acquisition metadata attached. Peak-list support requires no optional dependencies, whereas
 87 the vendor backends are optional and Thermo RAW support requires a .NET runtime.

88 Fragment matching accepts fragments from the companion peptacular library, which supports
 89 ProForma notation (LeDuc et al., 2022) and the current 2.1 specification. paftacular labels
 90 matched peaks in `mzPAF` notation. The resulting matches can be evaluated with the hyperscore
 91 popularized by X!Tandem (Craig & Beavis, 2004), the normalized spectral contrast angle
 92 (Toprak et al., 2014), matched fraction, and mass-error statistics. For collaboration and
 93 troubleshooting, `spectrl` tokens allow a specific spectrum to be shared in an issue or message
 94 without attaching a file.

95 Optional bidirectional adapters connect `spxtacular` spectra with `matchms Spectrum` and
 96 `spectrum_utils MsmsSpectrum` objects. The `matchms` adapter uses a namespaced payload to
 97 preserve the richer metadata available in `spxtacular`. Because the `spectrum_utils` representation
 98 is narrower, its adapter warns when fields cannot be represented. Dependencies are imported
 99 only when an adapter is called.

100 Research impact statement

101 `spxtacular` provides the spectrum processing and visualization behind the public Spectra web
 102 application, an interactive spectrum viewer. In our laboratory, it is used to analyze and visualize
 103 output from the Sage search engine (Lazear, 2023) and, in ongoing unpublished work, in a
 104 post-translational modification motif ratio quantitation workflow for covalent protein painting
 105 data (Son et al., 2023) studying transthyretin amyloidosis (ATTR); preliminary results from

that study were presented at ASMS 2026 (Pankow, 2026). spxtacular was also demonstrated alongside peptacular and related applications in an ASMS 2026 poster (Garrett et al., 2026).

AI usage disclosure

Anthropic Claude Code with Opus and Fable 5, and OpenAI Codex with Sol, were used for code editing and manuscript review. The authors made the core design decisions and reviewed, edited, and validated all AI-assisted outputs. They retain responsibility for the software and manuscript.

Author contributions

Patrick T. Garrett designed and implemented the library and drafted the manuscript. John R. Yates III directed the project and revised the manuscript.

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Conflicts of interest

The authors declare no competing financial interests.

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